# D2 shadow plenum structural supplement

Compact Kowa review | Corrective documentation 12 October 2026 | Separate from the historical structural report

The selected candidate changes the planning mass by **+38.88 g** relative to export18. J4 floor moment changes by **+5.48%** and conservative grip moment by **+4.40%**. The proposed compact equal-share J4 coupon screen is **0.95 N m**; the all-variant screen is **1.00 N m**. These are analytical screening loads, not qualified capacities.

**PRINT AND POWER HOLD.** No FEA, drop rating, print test, adhesive qualification or physical strength result is claimed. All 56 inherited physical gates and M1 remain open. All four plenum mouths, roof layers and bonded catch require physical evidence.

## Planning mass and load comparison

| Quantity | Reviewed export18 | Selected candidate |
|---|---|---|
| Complete mass | 831.38 g | 870.26 g |
| Tub / panel mass | 122.58 / 37.41 g | 155.72 / 38.19 g |
| Upper-body mass | 640.43 g | 679.12 g |
| Upper-body 5g floor moment | 1.7201 N m | 1.8144 N m |
| Conservative grip 5g moment | 4.4751 N m | 4.6720 N m |
| Carrier payload | 122.60 g | 127.27 g |

Estimated complete CoM: (-46.47, 1.02, 29.96) mm. Core: 158 x 63.6 x 100 mm. Manifest mesh-derived bounds: 248.83 x 72.82 x 216.82 mm. These are CAD/proxy estimates, not a weighed or measured camera.

## Free body and retained sections

Changed manifest rows: s_c4, base_grip, eyecup, carrier, tub, panel, lens_collar. All other rows are identical. This is metadata consistency, not a new geometric equivalence test. Retained local sections require source/STEP preservation evidence. Old full-tub and full-panel zero-difference bindings do not apply.

**Upper-body method:** new free body: complete manifest minus the J4 lower rows measured from their bound STEPs (lower rows changed: base_grip). The lower grip and battery are excluded from J4 upper-body loading. Grip screening places the entire assembly mass at the upper-body CoM, with hand reaction at Z=-55 mm. The 1g/3g/5g cases are quasi-static handling sensitivities.

Equal-share J4 nominal lip/head stresses are 8.98/9.92 MPa; 30-70% station sharing scales them by 0.6-1.4. Grip gross nominal stresses are 1.59/1.69 MPa. The carrier screen is 3.52 MPa and 0.414 mm at E=1600 MPa and half the 5g payload reaction. Where carrier mass changes, these retained-arm results scale with payload. Contact, root notches, screws, slip, wear, anisotropy, warm creep and service pulls remain unqualified.

## Coupon load coverage

The retained 12 g wiring-location sensitivity at Z=100 mm gives 0.9366 N m per equally sharing compact joint. The old 0.90 N m target does not cover that case. A 70% station reaction gives 1.3112 N m and remains a separate unqualified case.

| Variant | Equal-share + wiring | 70% station reaction |
|---|---|---|
| compact-fujinon hf6xa | 0.7698 N m | 1.0777 N m |
| compact-kowa lm6hc | 0.9366 N m | 1.3112 N m |
| original-kowa lm6hc | 0.9820 N m | 1.3748 N m |

The rounded all-variant equal-share screen is 1.00 N m, governed by original-kowa_lm6hc. Fixture capacity and procedure need review before an inert-coupon test. C08/C09 omit the rear joint, independent lock and full tub; a local result cannot establish a whole-camera rating.

## Plenum geometry and physical print holds

shadow-gap plenum trial4 (windowed flow ribs, supported stepped front mouth, cold-mouth lintels, skin bed chamfer). The skin is 1.6 mm, the gap 6 mm and the inner wall 2.5 mm. The wall becomes a roof in the tub's -Y-down print pose. Keep the 0.20 mm layer baseline and declared orientations as starting conditions. Geometric bridge checks do not prove slicer direction, first-layer continuity, pillar bonds, sag or debris removal. Trapped automatic supports are unacceptable.

The 3 source-bound open-edge plenum coupons are small roof/beam/pillar crop (legacy regular-grid name): 48.0 x 32.0 x 10.1 mm (valid saved-STEP single solid); port-service and edge crop: 110.0 x 100.0 x 10.1 mm (valid saved-STEP single solid); all-four-mouths crop: 158.0 x 100.0 x 10.1 mm (valid saved-STEP single solid). They remain separate from the 9 printed assembly rows (10 physical production prints, because the tub body and bonded stub print separately), the centring gauge and the nine retained structural coupon pieces. None has been sliced, printed or tested. The all-four-mouths crop opens the inner wall face and omits the tub body; cut edges cannot establish closed-body cleanout or tub stiffness. The smaller crops omit regions of the complete plenum. Coupon geometry does not prove roof toolpaths, strength, pressure loss, thermal performance or a whole-camera rating.

## Measured area and occlusion scope

| Saved-STEP geometric quantity | Area | Reported status |
|---|---|---|
| cold path min cut | 420.00 / 420.00 mm2 | PASS |
| hot path min cut | 510.00 / 510.00 mm2 | PASS |
| Hot parallel branch series screen | 575.20 / 510.00 mm2 | PASS |

Path minimum cut is a geometric graph-capacity screen over the saved tub, not a fan operating point or measured flow. The series screen samples named sections; neither gives pressure loss or temperature. A PASS here closes only its stated geometric rule.

Blocking the hot underside mouth leaves the front mouth, whose minimum sampled saved-STEP neck is 274.20 mm2 against 510.00 mm2 of measured hot wall ports, before internal losses. The 591.00 mm2 sum of the two sampled mouth necks cannot be credited in that case. These areas are geometric samples, not useful airflow or rated blocked-vent operation.

Test unobstructed operation, normal hand positions, fitted straps and stand/support contacts separately. External bay separation excludes the main cavity and does not seal the baffle, CSI bypass or grip communication. Actual cooler registration, lead sweep, complete wiring, useful flow, SSD temperature and closed-body battery temperature remain open.

## Materials bonds and ribbon evidence

The selected manifest records 10 mm bonded engagement with a 3.1 x 1.6 mm tenon section, X/Y/Z pocket clearances 0.1/0.2/0.1 mm, ceiling top Z95.2 mm and 1.6 mm ceiling floor. These are modeled dimensions. The existing datum-tilt case and catch-window criterion are retained; current geometric audit results must be read separately from bond, torque and creep qualification.

The current full main audit records 130/130 passing bonded-stub checks. Its minimum reported tilted-window clearance is 0.106 mm against the unchanged 0.100 mm criterion, giving 0.006 mm nominal headroom. This small modeled margin does not establish tolerance-robust manufacture. Print variation, seating, adhesive film/cure, datum movement and warm creep remain physical HOLDs.

A separate 2-piece bonded-catch study is generated from the selected tub print stock and the exact production insert: catch-host-coupon: 20.0 x 24.0 x 63.6 mm, -Y down; catch-insert: 6.3 x 12.0 x 15.0 mm, -Z down. It retains production layer orientations. It screens local seating, datum tilt, adhesion and warm creep only. Cropped host and fixture omit whole-tub compliance and complete lens/camera constraints, so no whole-tub 0.5 N m torque qualification follows. The study is additional to the retained structural and plenum coupons, with no production-part count change. It has not been sliced, printed, bonded or mechanically tested.

The mass model retains ASA 1.07 g/cm3 and the empirical 0.85 factor. Polymaker's reference density is 1.13 g/cm3: **+5.6% to the ASA contribution** at fixed volume and factor. This is not the same percentage increase in complete-camera mass, and no new grade or density is adopted. Actual print and harness weights are needed before final CoM and load qualification.

Retain E=0.8/1.6/2.4 GPa and 5/10/20/30 MPa strength scenarios as sensitivities only. Published tensile results and HDT do not establish D2 allowables, warm creep, loaded-service temperature or a safety factor. Exact filament grade/process, epoxy SKU/preparation/bondline/cure and the actual camera, HDMI and OLED ribbon minimum bend radii remain unknown.

A deeper bonded engagement may improve the geometric tolerance screen while retaining its datum-tilt test; it still needs production-pose coupons with the selected epoxy, hot dwell and measured permanent shift. No supplier ribbon bend limit is inferred from a different cable family. Confirm seating, latch access, strain relief, continuity and full-mode operation on the actual routed parts.

## Source identity and research applicability

Calculation: concealed-vent-structural.json. Mouths: shadow-plenum-checks.json. Path and series areas: check-scope-checks.json. Bonded-stub rows: the current full verification.json. Coupon dimensions and validity: the current coupon index and its individual receipts. The machine-readable PDF receipt binds each source hash and records visual review separately.

The 12 October materials/ribbons and cooler research describe preserved baseline bdfd44fc and identify evidence gaps. The accompanying RESEARCH-INTEGRATION.md maps their conclusions to this corrective review; baseline geometry figures are not carried forward as current measurements. Manufacturer sources include Polymaker ASA data, Raspberry Pi camera instructions, Adafruit 3562 and the Active Cooler brief.

Selected mechanical manifest

`294220bc7e1babc507e00d41542d108bfbf172250e45f20ba828e83437634ae4`

Frozen export18 reference manifest

`c9e1e20710bb5f947ad47fa62328cebffb099fd3832d22909056bf8f1867cc76`